



Creating and Running Apps on the Microsoft Azure Hybrid Cloud



Microsoft Azure is a cloud computing platform developed to build, deploy and manage Web applications through cloud services. It provides both PaaS and IaaS services. Besides Microsoft software, it supports various programming languages, tools, frameworks and other third party systems and software.

Cloud computing is currently one of the key domains of research. It is used for the delivery of assorted computing services including telecommunications, storage, platforms, software, virtual infrastructure and many others. Cloud computing offers the delivery of virtual resources without the need to worry about establishing and maintaining the actual infrastructure. The end users of cloud services access the remote computing resources using Web based clients without having very complex configurations in their own devices. Those with just 1GB RAM in their own devices can work on the 128GB RAM installed in the cloud server without any issue. In this case, the system or device of the user acts as the terminal that accesses and works on the remote cloud infrastructure.

Cloud computing delivery channels

Cloud computing services are delivered by the service providers using different specific models, as listed below.

Infrastructure as a Service (IaaS): The IaaS cloud involves the delivery of computing infrastructure such as a virtual machine disk image library, raw block storage, object storage, firewalls, load balancers, IP addresses, virtual local area networks, etc, on demand, from the service providers' vast pool of computing resources installed in data centres.

Platform as a Service (PaaS): The PaaS models involve

cloud service providers delivering a complete computing platform, including operating system, a programming language execution environment, an integrated development environment, database server, Web server and other related technologies. Popular PaaS implementations include Microsoft Azure and Google App Engine.

Software as a Service (SaaS): In SaaS, also known as on-demand software, the cloud users are provided access to application software and databases. Cloud providers manage the infrastructure and platforms that run the applications.

Metal as a Service (MaaS): This is a provisioning construct developed by Canonical, the developers of Ubuntu, to assist and integrate the deployment and dynamic provisioning of hyperscale computing environments such as Big Data workloads and cloud services, in a dynamic manner.

Network as a Service (NaaS): This service enables the end user to create the virtual network on the cloud without physical installation using hubs, switches and other physical resources. Using this approach, any number of computers can be attached and worked on at remote data centres.

Hybrid cloud services

Hybrid cloud services enable people to use different types of cloud services simultaneously, with just a single point